BIOCHEMISTRY CUMULATIVE EXAM

March 7, 1992

The glycolysis reaction:

Fructose-6-phosphate + ATP → Fructose-1,6-bisphosphate + ADP

is catalyzed by the enzyme phosphofructokinase. The gluconeogenesis reaction:

Fructose-1,6-bisphosphate + H_2O → Fructose-6-phosphate + P_i

is catalyzed by the enzyme fructose-1,6-bisphosphatase. A new mutant rat has been discovered that is unusually sensitive to hyperthermia (overheating). Under normal conditions the rat is OK, but if it exercises vigorously, the rat's body temperature increases rapidly without control. The mutant rat contains a single mutation in the gene for fructose-1,6-bisphosphatase that is responsible for the hyperthermia.

1. (50) Discuss how the activities of the enzymes phosphofructokinase and fructose-1,6-bisphosphatase are regulated in normal rats. (If you cannot remember the details of this, present a logical scheme for how these enzymes could be regulated that would account for the regulation of glycolysis and gluconeogenesis).

2. (50) Discuss how the regulation of these enzymes is altered in the mutant rat. What triggers the hyperthermia episode in the rat, and where does the heat come from?